PADHAKU
Curiosity · Class 6 Science



CHAPTER 9

Methods of Separation in Everyday Life

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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⭐ Handpicking, threshing and winnowing

? **They always stay in touch with their relatives and friends. Do you also keep in touch with your loved ones?**

> 💡 **SAMPLE ANSWER**
>
> Yes. I talk to my grandparents and cousins on the phone and visit them during holidays and festivals.

Have fun while you learn **Explore folk songs of your region and try to sing along with your friends.**

> 💡 **SAMPLE ANSWER**
>
> Ask elders at home for folk songs sung at harvest time, weddings or festivals in your region and sing them together with friends.

? **A small amount of puffed rice is mixed with chana dal. Can you think of separating the mixture by any method other than handpicking?**

> ✔ **ANSWER**
>
> Yes.
>
> - **Winnowing:** puffed rice is very light, so it is blown away by air, while the heavier chana dal falls down.

- **Sieving:** puffed rice grains are bigger, so they stay on a sieve whose holes let the smaller dal pass.

Activity 9.1 **Take a handful of roasted peanuts and rub them between your palms. What happens?**

Is it possible to separate the removed skin and the peanuts?

Now, try blowing it. What do you observe? Which of these components —removed peanut skins or peanuts are blown away?

How do you think farmers separate so many grains from husk?

✔ ANSWER

The thin red skins come off the peanuts.

Yes, they can be separated.

On blowing, the **light peanut skins are blown away** and the heavier peanuts stay in the hand.

Conclusion: Blowing air separates heavier and lighter components.

Farmers use the same idea: they drop the grain–husk mixture from a height in the wind (using a soop). The wind blows the light husk away and the heavy grains fall down.

Fig. 9.4 **What do you infer from Fig. 9.4? Do both the components, wheat grains and husk,**

fall at the same place? Which of the two components get blown away? Can the wind separate the two components?

✔ ANSWER

No, they do not fall at the same place.

The heavier **wheat grains fall straight down** near the farmer, while the lighter **husk is blown away** by the wind to a farther place.

Yes, the wind can separate the two components. This is called **winnowing**.

? **Valli is unable to separate husk from rice in a closed room. How can you help her?**

✔ ANSWER

In a closed room there is no wind. She can switch on a **fan** (or use a hand fan / blow air) and then winnow the rice, or take the rice outside where the wind is blowing.

? **Have you seen any similar kind of activity being done at your home?**

💡 SAMPLE ANSWER

Yes. My mother uses a soop to toss rice or dal; the light husk and dust fly out and the grains stay in the soop.

⭐ Sieving

? **Carefully observe a sieve. Are all the holes of the sieve the same size?**
Will sieving work if the holes of the sieve are larger than the substances?
Is there any difference in the size between the particles that pass through the sieve and the particles that remain on the sieve?

✔ ANSWER

Yes, in one sieve all the holes are of about the **same size**.

No. If the holes are larger than all the substances, everything will pass through and nothing will be separated.

Yes. Particles **smaller** than the holes pass through; particles **bigger** than the holes (like bran and small stones) remain on the sieve.

? **Have you ever observed sieves being used at construction sites to separate pebbles and stones from the sand?**

✔ ANSWER

Yes. Workers throw sand on a large slanting wire-mesh sieve. Fine sand passes through, and pebbles and stones stay back or roll down.

More to do! **What is Sabarmati Ashram famous for? Draw a poster showing Dandi March**

and discuss why it was organised.

💡 SAMPLE ANSWER

Sabarmati Ashram in Ahmedabad was the home of **Mahatma Gandhi**. The **Dandi March** (Namak Satyagrah) started from here in 1930.

It was organised to protest against the British law that put a tax on salt and did not allow Indians to make their own salt. Gandhiji walked to Dandi on the seacoast and made salt from seawater.

Poster: draw Gandhiji with a stick leading people walking towards the sea, with the words 'Dandi March 1930'.

⭐ Evaporation

More to do!

Find out about some water bodies in India that contain common salt. One such source is Sambhar Lake in Rajasthan.

💡 SAMPLE ANSWER

- Sambhar Lake, Didwana Lake and Pachpadra (Rajasthan)
- Chilika Lake (Odisha) and Pulicat Lake (Andhra Pradesh–Tamil Nadu), which have salty (brackish) water
- The Rann of Kutch (Gujarat) and the sea along the coast of India

Activity 9.2 **Dissolve 2–3 teaspoons of salt in half a bowl of water. Spread a few drops of the salt solution on black or dark coloured thick paper and allow it to dry.**

Do you observe some patches on the paper? What do you think is left on the paper? Where has the water disappeared?

✔ ANSWER

Yes, **white patches** appear on the paper after it dries.

Salt is left on the paper (we can feel it by touching).

The water has **evaporated** — it changed into water vapour and went into the air.

? **Have you ever observed white patches on the dark coloured clothes you wear during hot summers? How are these patches formed?**

✔ ANSWER

Yes. In summer we sweat a lot. Sweat has salt dissolved in water. The water of the sweat **evaporates**, and the **salt is left behind** as white patches on the clothes.

Activity 9.3 **Heat some salt solution in a china dish and let the water boil away. Allow the**

dish to cool down.

What do you observe? What is left in the china dish? Did you get the salt back?

✔ ANSWER

The water slowly **evaporates** (boils away) as steam.

A **white solid — salt** — is left in the china dish.

Yes, we get the salt back. **Conclusion:** A solid dissolved in a liquid can be separated by **evaporation**.

? **Is there any method through which I can get back both salt and water?**

Observe the figure. Does it answer the question? Can you name the process involved?

✔ ANSWER

Yes. Heat the salt solution and hold a **cold plate or lid** above it. The water vapour cools on the plate and changes back into water drops, which can be collected. Salt is left in the vessel.

The processes involved are **evaporation** followed by **condensation**.

⭐ Sedimentation, decantation and filtration

? **Oil does not mix with water and forms a separate layer when left undisturbed for some time. Which method of separation would you use to separate oil and water?**

✔ ANSWER

Decantation. Leave the mixture undisturbed so the oil forms a layer on top. Then gently tilt the vessel and pour out the oil into another vessel.

? **Can Malli use a tea strainer to filter muddy water? Also, try to filter the muddy water through a piece of cloth and observe the difference.**

✔ ANSWER

No. The holes of a tea strainer are too big, so the tiny mud particles pass through it.

A piece of cloth has much smaller pores between its threads, so the water that comes out is **clearer**.

? **How many layers of cloth do I need to use to get clear water?**

💡 SAMPLE ANSWER

The more layers (folds) of cloth we use, the smaller the gaps, and the clearer the water. Folding a thin cloth 4 to 8 times usually gives fairly clear water. If it is still muddy, use a filter with even finer pores, like **filter paper**.

Activity 9.4 **Fold a filter paper to make a cone, place it in a funnel kept on a conical flask and pour muddy water into it. What do you observe? Do the mud particles pass through the filter paper?**

✔ ANSWER

No, the mud particles do not pass through the filter paper because its pores are very fine.

Mud is left on the filter paper as **residue**, and **clear water** collects in the conical flask as **filtrate**.

Activity 9.5 **Valli collects some pond water with unwanted substances in it. Design and create a working model of a water filter using low-cost materials.**

💡 SAMPLE ANSWER

- Cut the bottom of a plastic bottle and keep it upside down (cap side down, cap removed) in a glass.
- Put layers from bottom to top: **cotton or cloth**, **charcoal**, **fine sand**, **coarse sand**, **small pebbles/gravel**.

- Pour the pond water from the top. The layers trap dirt; clearer water drips into the glass.

Note: this water is clearer but not yet safe to drink; it must be boiled first.

Poem **Let us raise awareness about river and ocean pollution by composing a poem. Add more lines to the poem 'Piece of plastic in my neck...'.**

💡 **SAMPLE ANSWER**

Bottles and wrappers fill our home,
Nowhere safe for us to roam.
Throw your waste in the bin, my friend,
Let this sorrow come to an end.

Discuss with your parents **A bowl of milk in your home has gone sour. Discuss with your parents how you can use it in another way. Also, which method of separation will you use in the process?**

💡 **SAMPLE ANSWER**

We can make **paneer (chhena)** from it. Heat the sour milk so that it curdles, then pour it through a **muslin cloth**. The solid paneer stays in the cloth and the liquid (whey) passes through.

Method used: **filtration** (the whey can also be poured off by **decantation**).

⭐ Churning and magnetic separation

? **Can you name one kitchen appliance which runs on electricity that is used to prepare buttermilk?**

✔ ANSWER

A **mixer (mixer-grinder) or blender**.

? **The children asked the carpenter to move a magnet through the sawdust and all the nails got attracted to it. Which method of separation did the carpenter use?**

✔ ANSWER

Magnetic separation. Iron nails are magnetic, so they stick to the magnet; sawdust is non-magnetic and is left behind.

Activity 9.6 **Put each slip in one of two baskets: A — To remove any one of the components that is not useful; B — To separate two different useful components.**

1. Small stones from pulses 2. Churning curd to obtain butter 3. Green chillies from cooked dalia or poha 4.

Seeds from watermelon 5. Sawdust and iron nails from building material 6. Marigold flowers from other flowers for a garland 7. Pebbles from sand 8. Coconut pieces from rice flour 9. Oil from water 10. Salt from salt solution

💡 **SAMPLE ANSWER**

Slip	Basket
1. Stones from pulses	A (stones not useful)
2. Churning curd	B (butter and buttermilk both useful)
3. Green chillies from dalia/poha	A
4. Seeds from watermelon	A
5. Sawdust and iron nails	B (both can be used)
6. Marigold from other flowers	B (other flowers can also be used)
7. Pebbles from sand	B (both used in building)
8. Coconut pieces from rice flour	B
9. Oil from water	B (both can be used)
10. Salt from salt solution	A (we want only the salt; water is removed)

Some slips can go in either basket depending on the situation; give a reason for your choice.

Play a Game—WISE FISH

Fish out a red fish (method of separation, Tank 1) and then the blue slip (Tank 2) that goes with it.

✔ ANSWER

Tank 1 (method)	Tank 2 (matching slip)
Filtration	insoluble particles get filtered as residue
Decantation	mixture of oil and water
Condensation	conversion of water vapour into its liquid state
Handpicking	large particles are handpicked
Churning	extract butter from curd
Evaporation	separate salt from seawater
Winnowing	lighter component of mixture is separated by blowing air
Sedimentation	heavier particles settled down
Sieving	difference in size of solid particles
Threshing	beating stalks to remove grains
Magnetic separation	difference in magnetic properties

Let us enhance our learning

1 **What purpose does handpicking serve in the process of separation?**

(i) Filtration (ii) Sorting (iii) Evaporation (iv) Decantation

✔ ANSWER

(ii) Sorting. In handpicking we pick out and sort particles by hand on the basis of size, colour and shape.

2 **Which of the following substances are commonly separated using the churning method?**

(i) Oil from water (ii) Sand from water (iii) Cream from milk (iv) Oxygen from air

✔ ANSWER

(iii) Cream from milk. On churning, the lighter cream/butter separates and floats on top.

3 **Which factor is usually essential for the filtration?**

(i) Apparatus size (ii) Presence of air (iii) Pore size (iv) Temperature of the mixture

✔ ANSWER

(iii) Pore size. The pores of the filter must be smaller than the particles to be removed.

4 (i) **State True [T] or False [F] with reason: Salt can be separated from salt solution by keeping it under the Sun.**

✔ ANSWER

True. The Sun's heat evaporates the water and salt is left behind (as in salt pits by the sea).

4 (ii) **Handpicking should be used only when the quantity of one component is less.**

✔ ANSWER

True. Handpicking is convenient only when the particles to be removed are few (small quantity) and big enough to be picked by hand.

4 (iii) **A mixture of puffed rice and rice grains can be separated by threshing.**

✔ ANSWER

False. Threshing separates grains from stalks.

Correct: A mixture of puffed rice and rice grains can be separated by **winnowing** (or sieving).

4 (iv) **A mixture of mustard oil and lemon water can be separated by decantation.**

✔ ANSWER

True. Mustard oil does not mix with lemon water; it forms a separate layer on top and can be poured off gently.

4 (v) **Sieving is used to separate a mixture of rice flour and water.**

✔ ANSWER

False. Sieving separates solids of different sizes, and fine flour particles pass through a sieve with the water.

Correct: A mixture of rice flour and water can be separated by **filtration** (or sedimentation and decantation).

5 **Match Column I with Column II.**

Column I	Column II
(i) Gram flour mixed with black gram	(a) Handpicking
(ii) Chalk powder mixed with water	(b) Magnetic separation
(iii) Corn mixed with potatoes	(c) Decantation
(iv) Iron powder mixed with sawdust	(d) Sieving
(v) Oil mixed with water	(e) Filtration

✔ ANSWER

(i) → **(d) Sieving**

(ii) → **(e) Filtration**

(iii) → **(a) Handpicking**

(iv) → **(b) Magnetic separation**

(v) → **(c) Decantation**

6 **In what situations would you use decantation instead of filtration to separate solids from liquids?**

✔ ANSWER

- When the solid particles are **heavy** and settle down quickly at the bottom (e.g. washing rice and pulses, sand in water).
- When the quantity is large, or a filter is not available.
- When we do not need complete separation (like pouring tea without a strainer).

7 **Can you relate the presence of nasal hair to any separation process?**

✔ ANSWER

Yes, **filtration**. The hair in our nose works like a filter: it traps dust, dirt and germs from the air we breathe in, and lets clean air go inside.

8 **During the COVID-19 pandemic, all of us wore masks. Generally, what material are they made of? What is the role of these masks?**

✔ ANSWER

Masks are usually made of layers of **cloth (cotton)** or special non-woven fabric (as in surgical and N95 masks).

They work as a **filter**: their tiny pores let air pass but stop dust, droplets and germs (like viruses) from entering our nose and mouth.

9 **A mixture containing potatoes, salt and sawdust has been given to you. Outline a stepwise procedure for separating each component from this mixture.**

✔ ANSWER

1. **Handpicking:** pick out the potatoes (they are big).

2. Add **water** to the remaining salt and sawdust and stir. Salt dissolves; sawdust does not.

3. **Filtration:** filter the mixture. Sawdust stays on the filter paper (dry it); salt solution passes through.

4. **Evaporation:** heat the salt solution until the water evaporates; **salt** is left behind.

10 **Read the story 'Intelligent Leela', tick the most appropriate options and give a suitable title.**

Before her father felt thirsty/hungry, she went to the nearby pond to fetch some water/grains. ... the water was muddy and fit/unfit for drinking. ... she filtered/churned

the muddy water using a piece of paper/muslin cloth. Leela, then, cooled/boiled the water for about 10 minutes in a covered pan. After cooling/boiling, she filtered/churned it again and made it fit/unfit for drinking.

💡 SAMPLE ANSWER

thirsty, **water**, **unfit**, **filtered**, **muslin cloth**, **boiled**, **cooling**, **filtered**, **fit**.

Title: **Leela Makes Water Safe to Drink** (any suitable title).

📘 Learning further

Fun with parents

Under supervision of your elders, try to prepare some herbal remedies, for example tulsi kadha. Which methods of separation will you use while preparing herbal kadha?

💡 SAMPLE ANSWER

Tulsi leaves, ginger etc. are boiled in water. Then the kadha is poured through a strainer: **filtration** (or the leaves are allowed to settle and the liquid is poured off: **sedimentation and decantation**).

Stage play

Imagine you and your friend are Malli and Valli. Write dialogues of a play

presenting their entire 'Bharat ki Yatra', highlighting the methods of separation they observed. Enact it in your school assembly.

💡 SAMPLE ANSWER

Scenes: Haryana (handpicking, threshing, winnowing, sieving) → Ahmedabad (salt from seawater by evaporation) → Puducherry (sedimentation, decantation, filtration) → Bhopal (churning) → Shillong (magnetic separation).

Sample line — Valli: "Look, Malli! The wind is blowing the husk away and the grains are falling down. This is winnowing!"

Group activity

Observe and list separation methods you employed and noticed in your surroundings throughout a week. Explain the reasons and compile the ones used most. Compare with your group.

💡 SAMPLE ANSWER

Day	What I saw	Method	Reason
Mon	Straining tea	Filtration	remove tea leaves
Tue	Removing stones from dal	Handpicking	stones are few and big
Wed	Sieving flour	Sieving	remove bran

Thu	Washing rice	Decantation	pour off dirty water

Filtration and handpicking may be the most used.

Be a stalwart of your community

Interview a ragpicker(s) and prepare a case study about the method(s) of separation he/she uses in daily life. Encourage children below 14 to join a neighbourhood school.

💡 SAMPLE ANSWER

A ragpicker mostly uses **handpicking** to sort plastic, paper, glass and metal from waste, and may use a **magnet** to collect iron pieces. The sorted items are sold for recycling.

Tell children and their parents that education is free in government schools and help them get admission.

Be a reporter

(i) Gather newspaper clippings and articles on methods of separation used in your society, such as in fields or construction sites. (ii) Interview local farmers about the latest agricultural separation methods they use.

💡 SAMPLE ANSWER

(i) Collect and paste clippings on threshers, sieving sand at building sites, salt making, water filters etc.

(ii) Farmers may tell you about **threshers** and **combine harvesters**, which do threshing and winnowing together, and machines that clean and grade grains.

Think like a scientist

You are provided with a mixture of iron nails, sand, black pepper, stones, common salt and water. Which steps will you follow to separate each component?

💡 SAMPLE ANSWER

1. **Magnetic separation:** move a magnet through the mixture to take out the iron nails.

2. **Handpicking:** pick out the stones and black pepper (big particles).

3. **Sedimentation and decantation:** leave it undisturbed; sand settles at the bottom; pour off the salt water. Then **filter** it to remove the remaining sand.

4. **Evaporation:** heat the salt water; salt is left. To get the water back too, collect the vapour by **condensation**.

🎉 **Chapter 9 done! Revise the answers once more tomorrow.**